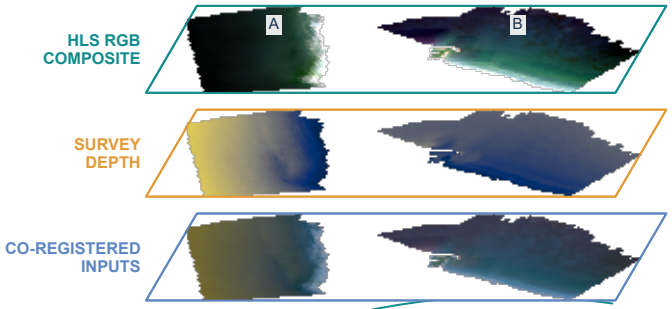


Satellite reflectance
HLS v2.0 · S30 Sentinel-2 + L30 Landsat 8/9
30 m · B02/B03/B04 + harmonized NIR

Bathymetric observations
XYZ soundings · 30 m cell medians
Anza 10 Nov 2022 · Ras El Ma 8 Aug 2024

Tide-model covariate
pyTMD + GOT5.6 overpass anomaly
Relative to survey-day mean; no local gauge tie

Pre-processing
HLS Fmask + common-band harmonization
XYZ projection + duplicate removal
30 m cell medians; retain local zeros

Sample construction
Quality-weighted four-band composite · ±30 d
Match cell depth + relative tide anomaly
Keep the two local vertical references

Models and validation
Stumpf · RF · HGB · RBF-SVR
Five spatial folds · 500 m blocks
90 m buffer · out-of-fold scores

SUPPORTED BATHYMETRY OUTPUT · RANDOM FOREST



Reported results
Spatially held-out RMSE, MAE and bias
Depth-stratified errors + block-bootstrap intervals
Tide, scene/sensor and cross-site sensitivity

Full-data maps are limited to 90 m from survey support; accuracy claims use out-of-fold predictions.

Left: Anza / Agadir · Right: Ras El Ma / Ras Kebdana
Each map keeps its local port zero.
EMODnet / GEBCO: source context only.